

Host Galaxy Metallicity Effects on Type Ia Supernovae Light Curves

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ABSTRACT

Within active star-forming regions, host galaxy metallicity is often used as a proxy for stellar metallicity of supernovae progenitor stars. In this paper, we present a study of the relationship between Type Ia supernova light-curve morphology and the metallicities (stellar and gas-phase) of their host galaxies. The morphology of each light curve is characterized by the change in magnitude 15 days after the peak (Δmag), providing a consistent basis of measurement applicable across our sample of 41 supernovae. Our analysis reveals no statistically significant correlation between host galaxy metallicities and Type Ia supernova light-curve morphology, in either the g-band or r-band. These results suggest that, within our sample, host galaxy metallicity does not play a dominant role in determining the light curve morphology of Type Ia supernovae, though it may instead influence other chemical characteristics of their progenitor environments.

1. INTRODUCTION

Host galaxy metallicity is commonly used as an estimate for the metallicity of Type Ia supernovae progenitor stars. It is suggested that host galaxy metallicity is the cause behind the relationship between Type Ia supernovae luminosities and stellar mass (Pan et al. (2014)). Furthermore, the luminosities themselves are correlated with host galaxy metallicities, with fainter supernovae in high metallicity galaxies as opposed to lower metallicity ones.

In the study done by Pan et al. (2014), host galaxy metallicity can reliably be used as a proxy for the metallicity of Type Ia progenitors. This is because the metallicity of these progenitor stars has a relatively show statistical agreement between the host galaxy metallicity and the small dispersion in actively star-forming galaxies. As such, the host galaxy metallicity should be similar. Passive galaxies, in contract, exhibit significantly larger dispersion in their Type Ia progenitor metallicities, therefore making the host galaxy metallicity unsuitable for predicting the metallicity of the supernova progenitors (Bravo & Badenes (2011)).

Understanding the relationship between host galaxy and progenitor metallicity is essential for our understanding of Type Ia supernovae and for reducing system uncertainties in cosmological distance measurements, as Type Ia supernovae are one of the cosmological standard candles. In this paper, we seek to investigate the relationship between Type Ia supernovae light curve morphology and the metallicity of their host galaxies (and by proxy, their progenitor stars) through a statistical analysis of an observational sample.

The two host galaxy metallicity values for this study are stellar metallicity and gas-phase metallicity. While the gas-phase metallicity can be aligned with the stellar metallicity of host galaxies, this relation is heavily dependent on galactic mass and radius, with the relation changing with these values (Zinchenko & Vílchez (2024)). Through Zinchenko and Vílchez (2024), we chose to study both metallicity properties in conjunction.

2. DATA COLLECTION

We collected Type Ia supernova data from the BLAST database, adding candidates to our catalog that had publicly accessible global (as opposed to local) galactic metallicity data. It was essential for candidates to have an assigned Zwicky Transient Facility (ZTF) identifier from the Transient Name Server (TNS) as the ZTF identifiers were necessary in obtaining light curves from the Automatic Learning for the Rapid Classification of Events (ALeRCE) project.

The data collection and visualization process was done in python throughout multiple files. The ‘GetData’ file contains a class that contains the names and data for each supernova observed, as well as methods on how to obtain that data. The ‘Files’ file contains methods on how to retrieve and store data in text files so that they can be quickly retrieved without having to query for the data each time it is needed. The ‘Lightcurves’ file handles anything with the light curve data, including converting apparent magnitude to absolute magnitude using redshift. The ‘Plots’ file handles plotting the light curve data, and the ‘OtherPlots’ file handles plots such as the metallicity vs. peak/delta brightness plots. Finally, the ‘PublicMethods’ file handles some common methods with proper comments and documentation, they include functions that retrieve all data, and plot light curves/other data.

When the get data function is initially called, it will query data from both BLAST and ALeRCE. When imputed the BLAST name, it will query a JSON file from the BLAST database using the python library urlopen to retrieve

it. From there, it will query TNS using the same library and use REGEX to search the web page for the ALerCE identification. Using that identifier it will then query ALerCE to retrieve any ALerCE data and the light curve data. Finally, using the RA and DEC from BLAST we pulled information on the extinction brightness using astroquery. This is how we obtained all of our data for this project.

In order to plot the light curves for each supernova we used the light curve data provided by the ALerCE database. We first corrected the magnitudes by converting the apparent magnitudes to absolute magnitudes. This was done using the redshifts of the supernovae and the WMAP9 object from astropy to convert a redshift into a distance, and a distance to convert apparent magnitude to absolute using the formula $M = m - 5\log_{10}(d) + 5$. The SDSS extinction magnitudes was pulled from IrsaDust within astroquery, and was then subtracted from the absolute magnitude in order to correct for external brightness factors. Using the corrected magnitudes and the time provided from the light curve data, we were able to plot both red and green detections and non detections, similar to what is shown on the ALerCE website but using our absolute magnitudes corrected for Milky Way extinction.

3. DATA ANALYSIS

3.1. Light Curves

We first plotted light curves for each supernova in our dataset using the corrected magnitudes and corresponding Julian dates. Both the SDSS r- and g-band detections, including their associated uncertainties, as well as non-detections, were plotted using values obtained directly from ALerCE. Vertical solid and dashed lines were included to mark the time of peak absolute magnitude and 15 days after the peak, respectively. As shown in Figure 1, many of the resulting light curves show typical Type Ia supernova behavior. However, several objects in our sample, including 2018hrt and 2018feb, show irregular features associated with the non-detection data. In addition, some light curves are incomplete and lack measurements at the 15 days marker. Examples of this include the light curves for 2022zwa and 2020bqz, for which no magnitudes were recorded at that time.

3.2. Metallicity

The metallicity data compares the peak absolute magnitudes and the change in brightness over 15 days (Δmag) for the supernovae in our sample with their host galaxy's logarithmic metallicities ($\log(Z/Z_{\odot})$), those being the stellar metallicity ($\log(Z_{*}/Z_{\odot})$) and the gas phase metallicity ($\log(Z_{gas}/Z_{\odot})$). The peak magnitude distributions show that most supernovae in the sample reach absolute magnitudes between -19 and -20, which is consistent with typical values for Type Ia supernovae. However, objects with $\log(Z_{*}/Z_{\odot})$ values in the range -0.5 to -0.25 appear to have a greater distribution in their peak magnitudes, including several atypical values that are both brighter and dimmer than the most in the data set. This is seen most clearly in Figure 2(a).

Many of the supernovae in our sample display Δmag values close to zero. There are some notable exceptions, with these outliers falling between -0.75 and -0.25 for $\log(Z_{*}/Z_{\odot})$. Despite these observations, the data contains significant noise, and any apparent trends may be driven by outliers within the relatively small sample size.

3.3. Statistical Analysis

To determine if there is an actual relationship between the supernovae light curve morphology and the metallicity of their host galaxies, we plotted the Δmag values for both the g- and the r-bands ($\Delta g\ mag$ and $\Delta r\ mag$, respectively) against $\log(Z/Z_{\odot})$. The majority of the stellar metallicities are less than solar metallicity, with only about 2 supernovae being in host galaxies that have stellar metallicities of the sun or slightly higher. For the gas phase metallicity, there are significantly more data points in host galaxies with gas phase metallicities higher than solar metallicity. In addition to plotting Δmag against $\log(Z/Z_{\odot})$, we color mapped the data points based on their redshift, as seen in Figure 3. Therefore, we must consider how the redshift impacts our statistical analysis. As such, we replotted Δmag against $\log(Z/Z_{\odot})$ in four different redshift bins.

Further analysis was done using the pearsonr package from SciPy. The pearsonr package takes in two variables, x and y, and subsequently calculates the Pearson correlation coefficient and the p-value (Virtanen et al. (2020)). The correlation coefficient is a normalized measurement of the covariance between two data sets, resulting in a number between -1 and 1. A correlation coefficient of -1 means that there is no correlation and a coefficient of 1 means that there is a correlation. It is calculated with the following formula using x, y, the average of x, and the average of y:

$$r = \frac{\Sigma(x - m_x)(y - m_y)}{\sqrt{\Sigma(x - m_x)^2 \Sigma(y - m_y)^2}}$$

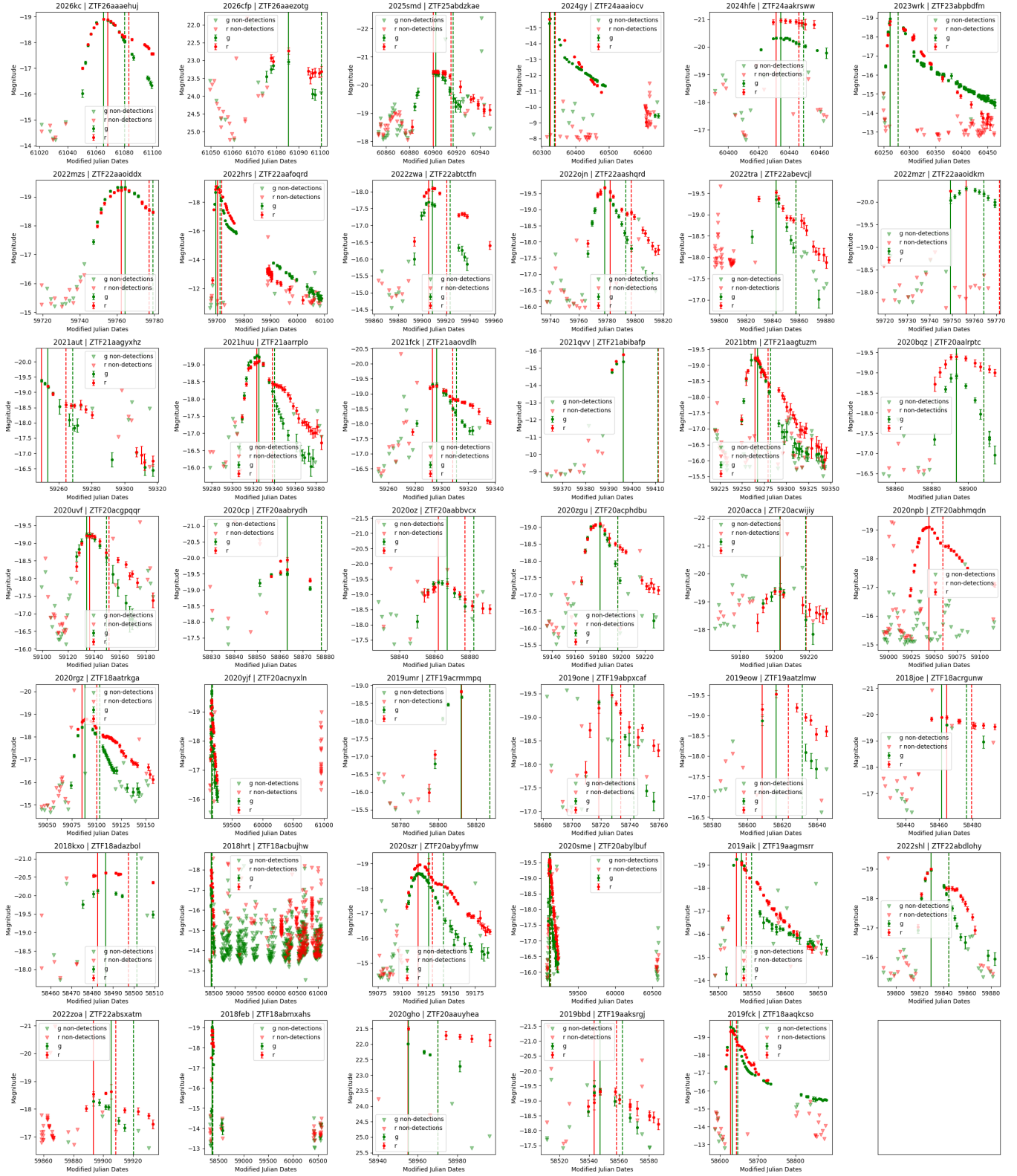


Figure 1. The light curves generated for the 41 Type Ia supernova in our sample, in both the g and r filters. The solid vertical lines in each light curve show when the absolute magnitude reached its peak, and the dashed lines show where the absolute magnitudes are 15 days after the peak.

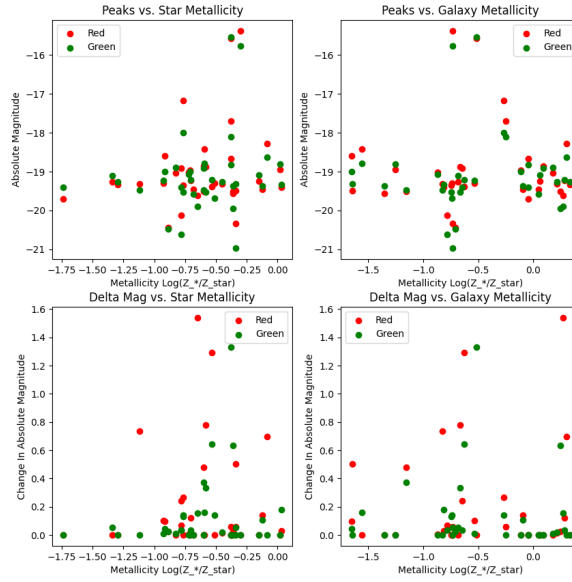


Figure 2. The peak magnitude and Δmag values for the 41 Type Ia supernova plotted against the logarithmic host galaxy metallicity, both stellar and gas phase. Panel (a) shows the peak absolute magnitude against the host galaxy’s stellar metallicity, panel (b) shows the peak against the host’s gas-phase metallicity, panel (c) shows the change in magnitude against stellar metallicity, and panel (d) shows the change in magnitude against gas-phase metallicity.

The p-value indicates the probability that uncorrelated data sets will reproduce the calculated correlation coefficient. P-values greater than 0.05 imply that the correlation is statistically insignificant. It is calculated by taking the continuous distribution of the negative absolute value of r and multiplying it by two. The following equation is that of the distribution function:

$$f(r) = \frac{(1 - r^2)^{\frac{n}{2} - 2}}{B(\frac{1}{2}, \frac{n}{2} - 1)}$$

With this package, we calculated the Pearson correlation coefficient and the p-value between a combination of the following datasets in our supernova sample: $\Delta g\ mag$, $\Delta r\ mag$, $\log(Z_*/Z_\odot)$, and $\log(Z_{gas}/Z_\odot)$. These values and the associated relationships are found in Table 1.

Table 1. Correlation between Δmag and Logarithmic Metallicity

Relationship	Pearson Correlation Coefficient	p-value
$\Delta g\ mag$ vs. $\log(Z_*/Z_\odot)$	0.1700	0.2817
$\Delta r\ mag$ vs. $\log(Z_*/Z_\odot)$	0.0214	0.8929
$\Delta g\ mag$ vs. $\log(Z_{gas}/Z_\odot)$	0.0331	0.8352
$\Delta r\ mag$ vs. $\log(Z_{gas}/Z_\odot)$	0.0349	0.8263

NOTE—Same Data as Figure 3

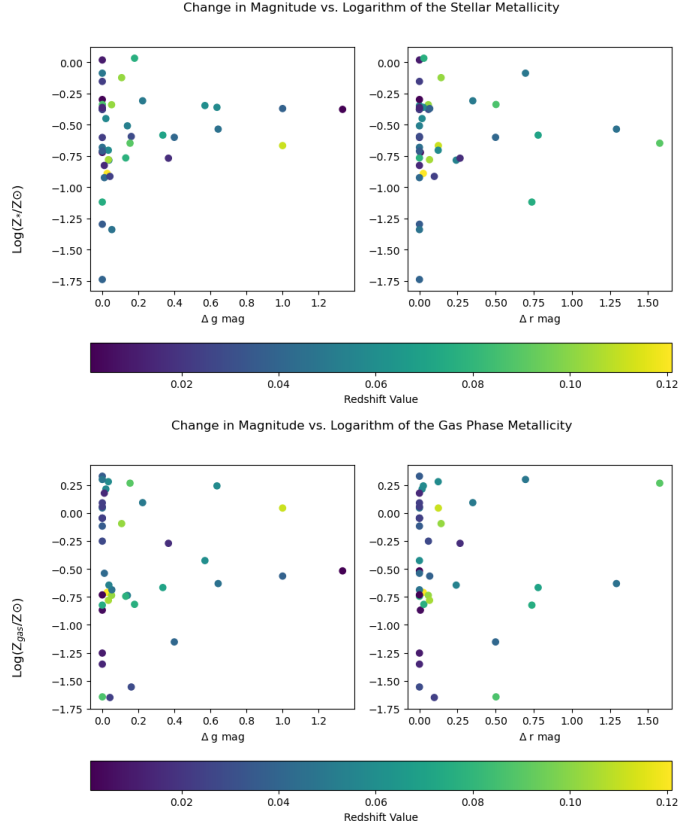


Figure 3. The Δmag values plotted in both the g- and r-bands against the logarithmic host galaxy metallicities, stellar and gas phase, while factoring the redshift of each transient for the color. The top panel (a) shows g and r Δmag against the logarithmic stellar metallicity. The bottom panel (b) shows g and r Δmag against the logarithmic gas phase metallicity.

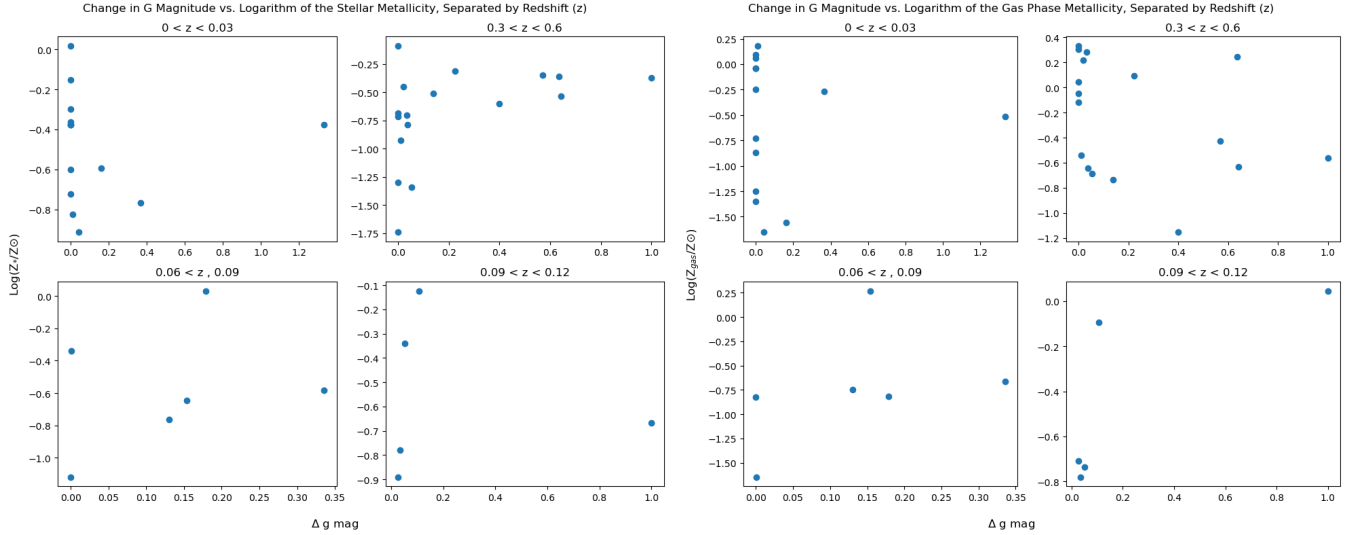


Figure 4. The Δmag for the SDSS g-band plotted against the logarithmic solar-scaled stellar and gas phase metallicities of the host galaxy, separated into four redshift bins. Redshift does not appear to be significant for the correlation between the change in magnitude for either the logarithmic stellar metallicity or the logarithmic gas phase metallicity for the SDSS g-band.

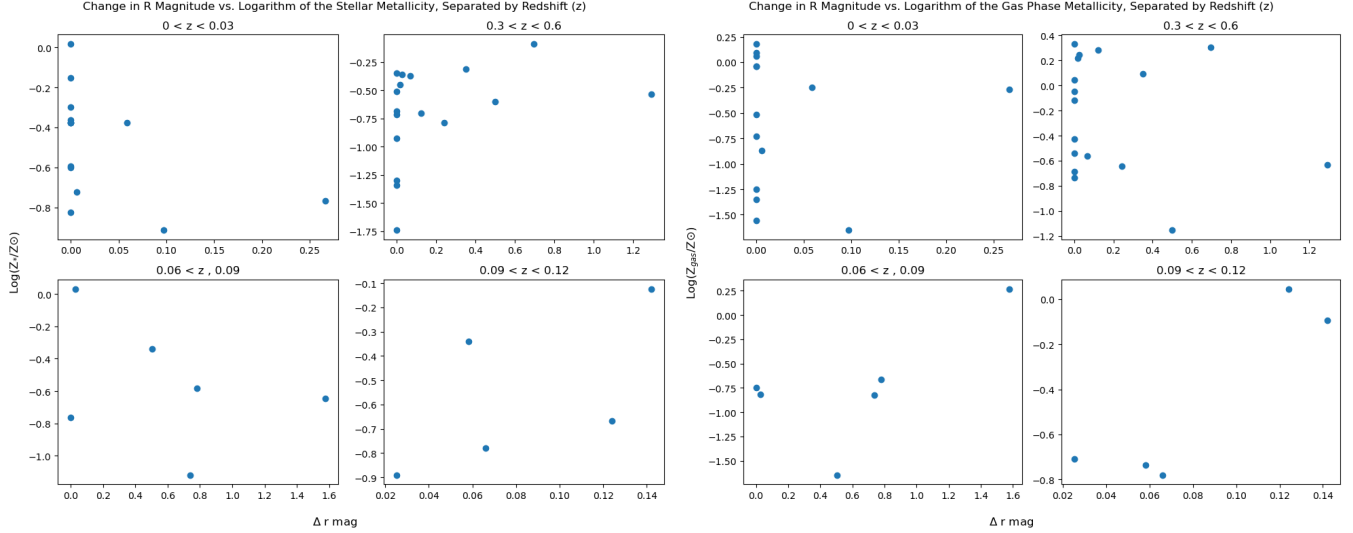


Figure 5. The Δmag for the SDSS r-band plotted against the logarithmic solar-scaled stellar and gas phase metallicities of the host galaxy, separated into four redshift bins. Redshift does not appear to be significant for the correlation between the change in magnitude for either the logarithmic stellar metallicity or the logarithmic gas phase metallicity for the SDSS r-band.

4. SUMMARY AND DISCUSSION

Our analysis shows no evidence for a significant statistical correlation between host galaxy stellar or gas phase metallicity and the light curve morphology of Type Ia supernovae.

From a visual inspection of Figures 2 and 3, we found that there is no significant correlation between the Δmag values and both the $\log(Z_*/Z_\odot)$ and $\log(Z_{gas}/Z_\odot)$ values. Figure 3(a), however, does appear to demonstrate a slightly stronger correlation than Figure 3(b), thus implying that the stellar metallicity of the host galaxy has more of an effect than the gas phase metallicity on the light curve morphology of Type Ia supernova.

This is corroborated by the calculated correlation coefficients and their corresponding p-values. As seen in Table 1, the strongest correlation coefficient of 0.1700 is associated with the relationship of $\Delta g mag$ and $\log(Z_*/Z_\odot)$. A correlation coefficient of 0.1700 displays a very weak positive linear relationship. Therefore, as the original stellar metallicity of the galaxy increases, the change in absolute magnitude 15 days after the peak of Type Ia supernovae increases slightly. However, the associated p-value for this correlation is 0.2817, which is more than five times greater than the significance limit of 0.05. As such, this weak correlation is extremely likely to be statistically insignificant.

The second strongest correlation coefficient is 0.0349, which is even weaker than the previous. It is associated with the relationship between $\Delta r mag$ and $\log(Z_{gas}/Z_\odot)$. While the p-value is even higher at 0.863, rendering this relation beyond insignificant, the existing relationship makes sense. In theory, there should be some sort of relation between the r-band magnitude and the gas phase metallicity because gas and dust within the universe is observed in the r- and i-bands.

The third strongest and second weakest correlation is between $\Delta g mag$ and $\log(Z_{gas}/Z_\odot)$, with a coefficient of 0.0331 and a p-value of 0.8352. As such, this relationship is incredibly weak and incredibly insignificant. The weakest correlation is between $\Delta r mag$ and $\log(Z_*/Z_\odot)$, with a coefficient of 0.0214 and a p-value of 0.8929. The lowest coefficient paired with the highest p-value suggests that there is effectively no relation between these two variables.

The plots displayed in Figures 4 and 5 show that the transient redshift has no apparent effect on the correlation between the magnitude and the metallicity as there are no general trends among the redshift bins.

There are several factors that may have affected the correlation, or lack thereof, between the change in magnitudes and the host galaxy metallicities. The first is our sample size. Our sample contained only 41 Type Ia supernovae, which is nowhere near representative of the true number of Type Ia supernovae that have been observed. Beyond that, we did not select specific bins for our data in the initial collection phase. It was quite difficult to determine what parameters we should have excluded or narrowed down within the scope of this project. This includes peak magnitude values, metallicity values, location in the sky, and specific redshift bins that actually correlate with an object's distance

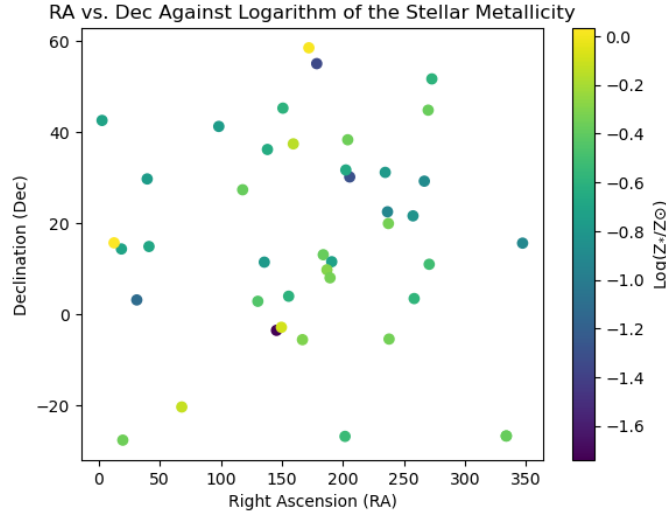


Figure 6. All 41 supernovae in our sample plotted in the sky, colored by their host galaxy’s logarithmic stellar metallicity (solar scaled).

in the sky rather than a later division. As seen in Figure 6, the supernovae in our data come from a wide range in the sky.

Other factors that affected our data analysis are the light curves themselves. In Figure 1, it is clear that not every supernova in our sample has a completed light curve, or even a readable one. However, we chose to include these within our analysis, which may have greatly skewed our results in the direction of non-correlation. The specific problem with the incomplete light curve is the lack of data exactly 15 days after the peak. Because of this, we were unable to reasonably obtain a reliable Δmag value for each supernova in our sample, not without interpolating the light curves themselves. We chose specifically not to interpolate these light curves because there may have been an odd peak at that specific 15-day marker, which would have then led to factually incorrect data, had we done some sort of linear interpolation. As such, there are many supernova in our data set that are said to have Δmag values of 0, which caused a heavy skew in the data set.

Although our study shows no correlation between host galaxy metallicities and light curve morphology for Type Ia supernovae, future studies may be done to verify or dispute this. These studies may look at supernovae in host galaxies with a specified metallicity range, in specific redshift ranges, in unique locations, or even in limited magnitude ranges. Other future studies may identify the relationship between Type Ia supernovae and other host galaxy properties, such as galaxy shape, galaxy size, or the dust parameters of the galaxy. And while we found no significant relation between host galaxy metallicity and light curve morphology, host galaxy properties (including metallicity) still significantly influence progenitor stars and the environments they are born out of. There is much to learn about Type Ia supernovae, their host galaxies, these resulting relationships, and what this tells us about Type Ia progenitor stars.

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